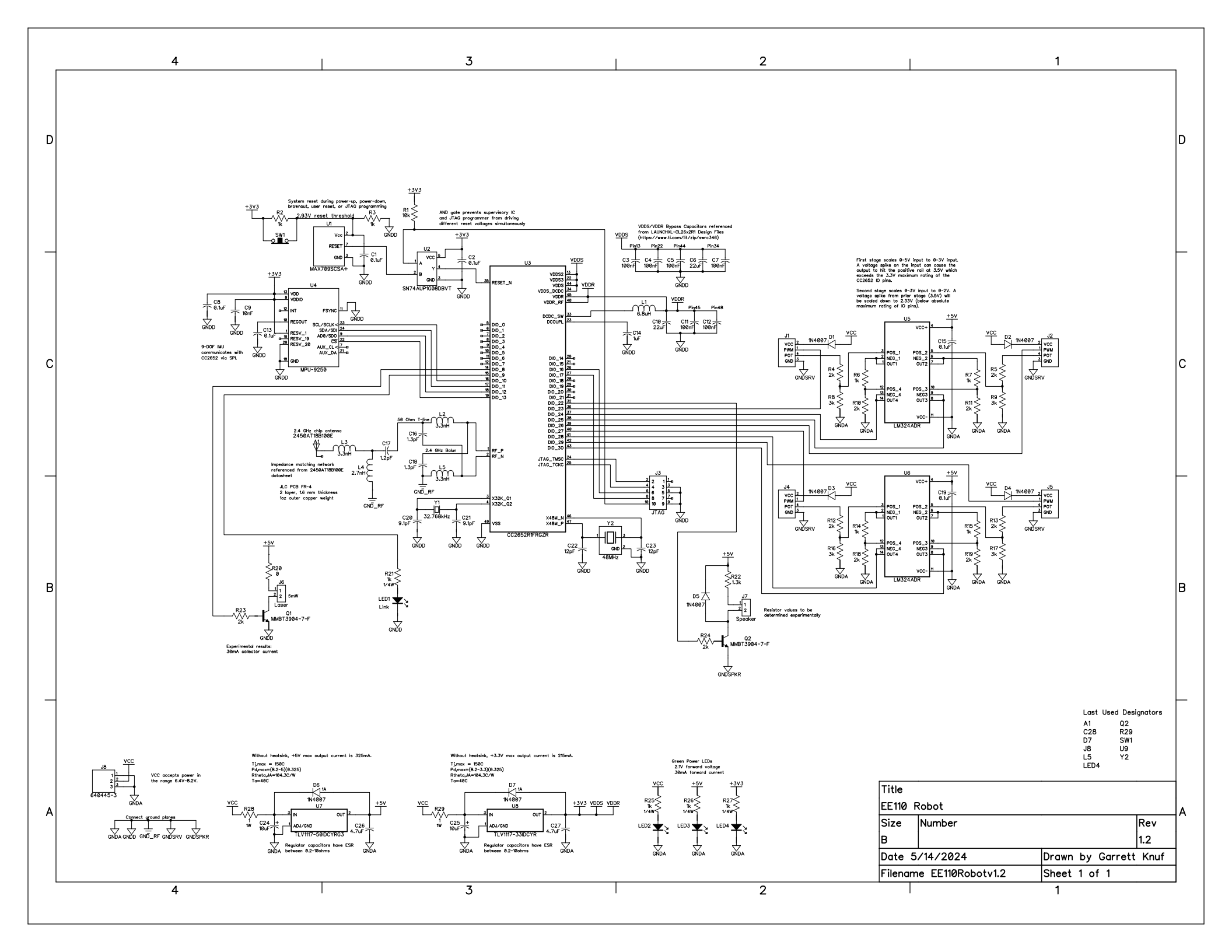

System reset during power-up, power-down, brownout, user reset, or JTAG programming
2.93V reset threshold
MAX709SCSA+
AND gate prevents supervisory IC and JTAG programmer from driving different reset voltages simultaneously
SN74AUP1G08DBVT
CC2652R1FRGZR
VDDS/VDDR Bypass Capacitors referenced from LAUNCHXL-CL26x2R1 Design Files (https://www.ti.com/lit/zip/swrc346)
9-DOF IMU communicates with CC2652 via SPI.
MPU-9250
First stage scales 0-5V input to 0-3V input. A voltage spike on the input can cause the output to hit the positive rail at 3.5V which exceeds the 3.3V maximum rating of the CC2652 IO pins.
Second stage scales 0-3V input to 0-2V. A voltage spike from prior stage (3.5V) will be scaled down to 2.33V (below absolute maximum rating of IO pins).
LM324ADR
2.4 GHz chip antenna 2450AT18B100E
50 Ohm T-line
2.4 GHz Balun
Impedance matching network referenced from 2450AT18B100E datasheet
JLC PCB FR-4 2 layer, 1.6 mm thickness 1oz outer copper weight
JTAG
Experimental results: 38mA collector current
Resistor values to be determined experimentally
Without heatsink, +5V max output current is 325mA.
Tj,max = 150C
Pd,max=(8.2-5)/(0.325)
Rtheta,JA=184.3C/W
Ta=40C
TLV1117-50IDCYRG3
Regulator capacitors have ESR between 0.2-10ohms
Without heatsink, +3.3V max output current is 215mA.
Tj,max = 150C
Pd,max=(8.2-3.3)/(0.325)
Rtheta,JA=184.3C/W
Ta=40C
TLV1117-33IDCYR
Green Power LEDs
2.1V forward voltage
38mA forward current
VCC accepts power in the range 6.4V-8.2V.
Connect ground planes
Last Used Designators
A1 Q2
C28 R29
D7 SW1
J8 U9
L5 Y2
LED4
Title
EE110 Robot
Size B
Number
Rev 1.2
Date 5/14/2024
Drawn by Garrett Knuf
Filename EE110Robotv1.2
Sheet 1 of 1